STATUS BOARD – Hadamard 14420 Paley-I (2026-10-09)

Cutoff: 2026-10-09T21:16:19Z (~5:16PM ET)

Ship

- Lab: `/workspace/experiments/band1/hadamard_14420_paleyI/`
- overall_pass=True; units 740/740; row blocks 113/113; wall_total=2514.1214s
- H_14420 Paley-I max_abs=0 wall=2476.098s q=14419
- H_14420 Paley-II FAIL/SKIP (q=7209=3?*89 NOT prime power; ===1 mod 4 but composite)
- H_14420 Kronecker FAIL/SKIP (4*3605; 3605===1 mod 4 NOT Hadamard-admissible; 3605=5*7*103 (odd composite))
- SPK: IN SPK via Paley-I q=14419 (verified)

Honesty

- clay_closed=false * GATE-04 LOCKED * conjecture OPEN * gap 2052 OPEN
- Skip: Miyamoto digression this wave (prefer classical SPK)
- Fold 14420 ONLY via classical Paley-I (never Miyamoto alone)

Next

q2=6961->n=13924 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6961 prime ===1 mod 4; Paley-I q=13923=3^2*71317 NOT prime -> FAIL; Kronecker 13924=4*3481; 3481=59^2 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3481); 13924=2^2*59^2 ===4 mod 16; needs a Paley-II wave (template hadamard_14356_paleyII / hadamard_9124_paleyII)) | q2=6977->n=13956 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6977 prime ===1 mod 4; Paley-I q=13955=5*2791 NOT prime -> FAIL; Kronecker 13956=4*3489; 3489=3*1163 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3489); 13956=2^2*31163 ===4 mod 16; needs a Paley-II wave (template hadamard_14356_paleyII / hadamard_9124_paleyII)) | q2=6997->n=13996 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=6997 prime ===1 mod 4; Paley-I q=13995=3^2*5*311 NOT prime -> FAIL; Kronecker 13996=4*3499; 3499 odd prime (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3499); 13996=2^2*3499 ===12 mod 16; needs a Paley-II wave (template hadamard_14356_paleyII / hadamard_9124_paleyII)) | q2=7001->n=14004 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7001 prime ===1 mod 4; Paley-I q=14003=11*1967 NOT prime -> FAIL; Kronecker 14004=4*3501; 3501=3*2389 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3501); 14004=2^2*3*2389 ===4 mod 16; needs a Paley-II wave (template hadamard_14356_paleyII / hadamard_9124_paleyII)) | q2=7013->n=14028 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7013 prime ===1 mod 4; Paley-I q=14027=13^2*83 NOT prime -> FAIL; Kronecker 14028=4*3507; 3507=3*7*167 odd (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3507); 14028=2^2*3*7*167 ===12 mod 16; needs a Paley-II wave (template hadamard_14356_paleyII / hadamard_9124_paleyII)) | q2=7057->n=14116 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7057 prime ===1 mod 4; Paley-I q=14115=3*5*941 NOT prime -> FAIL; Kronecker 14116=4*3529; 3529 odd prime (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3529); 14116=2^2*3529 ===4 mod 16; needs a Paley-II wave (template hadamard_14356_paleyII / hadamard_9124_paleyII)) | q2=7069->n=14140 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7069 prime ===1 mod 4; Paley-I q=14139=3^2*1571 NOT prime -> FAIL; Kronecker 14140=4*3535; 3535=5*7*101 odd (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3535); 14140=2^2*5*7101 ===12 mod 16; needs a Paley-II wave (template hadamard_14356_paleyII / hadamard_9124_paleyII)) | q2=7109->n=14220 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7109 prime ===1 mod 4; Paley-I q=14219=59*241 NOT prime -> FAIL; Kronecker 14220=4*3555; 3555=3^2*5*79 odd (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3555); 14220=2^2*3^2*5*79 ===12 mod 16; needs a Paley-II wave (template hadamard_14356_paleyII / hadamard_9124_paleyII)) | q2=7129->n=14260 Paley-II DIRECT (skipped by the Paley-I-only chain; q2=7129 prime ===1 mod 4; Paley-I q=14259=3*4753 NOT prime -> FAIL; Kronecker 14260=4*3565; 3565=5*23*31 odd (===1 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3565); 14260=2^2*5*23*31 ===4 mod 16; needs a Paley-II wave (template hadamard_14356_paleyII / hadamard_9124_paleyII)) | next Paley-I DIRECT after

*14420 (and FIRST any direct route after 14420): q=14423->n=14424 (prime ===3 mod 4; no prime in 14420..14422; Paley-II q2=7211 prime but ===3 mod 4 (needs ===1) -> FAIL/SKIP; Kronecker 14424=4*3606; 3606=2*3601 ===2 mod 4 -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3606); 14424=2^3*3601 ===8 mod 16) | FIRST Paley-II DIRECT after 14420: q2=7213->n=14428 (q2=7213 prime ===1 mod 4; Paley-I q=14427=3^2*7229 NOT prime -> FAIL; Kronecker 14428=4*3607; 3607 odd prime (===3 mod 4) -> NOT Hadamard-admissible -> FAIL/SKIP (do NOT invent H_3607); 14428=2^2*3607 ===12 mod 16; needs a Paley-II wave (template hadamard_14356_paleyII); comes after Paley-I 14424) | Kronecker-only (note-only): none in 14424..14428 (3606===2 mod 4, 3607 odd) (or gap 2052 / Track B)*